

RIPASSO 1

EQUAZIONI DIFFERENZIALI

$$y_k + a_{k-1} y^{(k-1)} + \dots = g(k) \quad g(k) = 0 \quad \text{OMOGENEA}$$

EQUAZIONI DIFFERENZIALI DI PRIMO ORDINE

$$\dot{y} = a(t)y + b(t)$$

SOLUZIONE

$$y(t) = e^{A(t)} \int e^{-A(t)} b(t) dt + c_0 \quad A(t) = \int_{t_0}^t a(\tau) d\tau$$

RISOLVO IL PROBLEMA DI CAUCHY

$$\begin{cases} \dot{y} = a(t)y + b(t) \\ y(t_0) = y_0 \end{cases}$$

$$y(t) = e^{A(t)} y_0 + \int_0^t e^{A(t)-A(\tau)} b(\tau) d\tau$$

$$a(t) = a \quad A(t) = a(t - t_0)$$

$$y(t) = e^{a(t-t_0)} y_0 + \int_0^t e^{a(t-\tau)} b(\tau) d\tau = e^{a(t-t_0)} y_0 + e^{at} * b(t)$$

SISTEMI DI EQUAZIONI DIFFERENZIALI

$$\begin{cases} \dot{x}_1 = a_{11}x_1 + \dots + a_{1n}x_n + b_{11}u_1 + \dots + b_{1m}u_m \\ y_1 = c_{11}x_1 + \dots + c_{1n}x_n + d_{11}u_1 + \dots + d_{1m}u_m \end{cases}$$

INGRESSO - STATO - USCITA

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases} \xrightarrow{x = Tz} \begin{cases} \dot{z} = \tilde{A}z + \tilde{B}u \\ y = \tilde{C}z \end{cases}$$
$$\begin{aligned} \tilde{A} &= T^{-1}AT \\ \tilde{B} &= T^{-1}B \\ \tilde{C} &= CT \end{aligned}$$

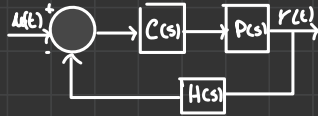
CONTROLLO A CATENA APERTA E CONTROLLO A CONTROREAZIONE

IL SISTEMA DI CONTROLLO A CATENA APERTA



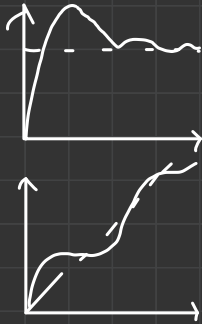
$$F(s) = C(s) \cdot P(s)$$

IL SISTEMA DI CONTROLLO A CONTROREAZIONE



$$W(s) = \frac{C(s)P(s)}{1 + C(s)P(s)H(s)}$$

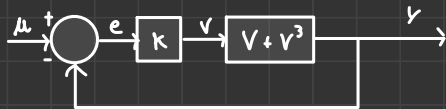
REGOLAZIONE E ASSERIMENTO



-RIGETTAZIONE DISTURBI

PROPRIETA' LINEARIZZANTE DELLA CONTROAZIONE

SISTEMA STATICO NON LINEARE



$$e = u - y \quad v = ke \quad v + v^3 = y$$

$$ke + (ke)^3 = y \rightarrow k(u - y) + k^3(u - y) - y = 0$$

$$\frac{k(u - y) - y}{k^3} + (u - y)^3 = 0 \quad k \rightarrow \infty$$

$$(u - y)^3 = 0 \rightarrow u - y = 0 \rightarrow u = y$$

SISTEMI DINAMICI NON LINEARI



$$u(t) = Md^2\ddot{\theta} + Mgd \sin(\theta)$$

$$u(t) = \frac{1}{\varepsilon} (v - \dot{\theta})$$

$$Md^2\ddot{\theta} + Mgd \sin(\theta) = \frac{1}{\varepsilon} (v - \dot{\theta})$$

$$\varepsilon (Md^2\ddot{\theta} + Mgd \sin(\theta)) = v - \dot{\theta} \quad \varepsilon \rightarrow 0 \rightarrow \frac{1}{\varepsilon} \rightarrow \infty$$

$$v - \dot{\theta} = 0 \rightarrow v = \dot{\theta}$$

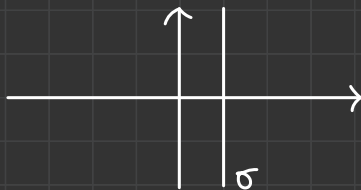
TRASFORMATA DI LAPLACE

$$\mathcal{L}\{f(t)\} = \int_0^\infty f(t) e^{st} dt \quad s = \sigma + j\omega$$

TRASFORMATE NOTEVOLI

$$\mathcal{L}\{e^{pt}\} = \frac{1}{s-p} \quad \mathcal{L}\{e^{j\omega t}\} = \frac{1}{s-j\omega}$$

$$\sin(\omega t) = \frac{e^{j\omega t} - e^{-j\omega t}}{2j} \rightarrow \begin{aligned} \mathcal{L}\{\sin(\omega t)\} &= \frac{\omega}{s^2 + \omega^2} \\ \mathcal{L}\{\cos(\omega t)\} &= \frac{s}{s^2 + \omega^2} \end{aligned}$$



$$\mathcal{L}\{e^{kt}\} = \frac{k!}{s^{k+1}}$$

$$\mathcal{J}_x(t) = \frac{t^{x-1}}{x-1} \quad \mathcal{L}\{k \mathcal{J}_x(t)\} = \frac{k}{s^x}$$

$\mathcal{J}_{-1}(t)$	$\mathcal{J}_{-2}(t)$	$\mathcal{J}_{-3}(t)$
1	t	$\frac{t^2}{2}$
$\frac{1}{s}$	$\frac{1}{s^2}$	$\frac{1}{s^3}$

$t > 0$
TRASF.

ANTITRASFORMATA DI LAPLACE

$$F(s) = \frac{N(s)}{D(s)} = \frac{\sum a_k s^k}{\sum b_k s^k}$$

DECOMPOSIZIONE IN POLI E RESIDUI

POLI SEMPLICI

$$F(s) = \frac{r_1}{(s-p_1)} + \frac{r_2}{(s-p_2)} + \dots$$

$$r_i = \lim_{s \rightarrow p_i} (s-p_i) F(s) \quad f(t) = r_1 e^{p_1 t} + r_2 e^{p_2 t} \dots$$

POLI MULTIPLI

$$F(s) = \frac{r_1^m}{(s-p_1)^m} + \frac{r_1^{m-1}}{(s-p_1)^{m-1}} + \dots + \frac{r_1}{(s-p_1)}$$

$$r_i^k = \lim_{s \rightarrow p_i} \frac{1}{(k-1)!} \frac{d^{k-1}}{ds^{k-1}} (s-p_i)^m F(s)$$

$$f(t) = \left(r_1^m, r_1^{m-1}, \dots, \frac{t^{m-1}}{(m-1)!} \right) e^{p_1 t}$$

POLI SEMPLICI COMPLESSI

$$F(s) = \frac{r}{(s-p)} + \frac{r^*}{(s-p^*)} \quad f(t) = R e^{(\sigma + j\omega)t} + R^* e^{(\sigma - j\omega)t}$$

$$R = |r| e^{j\varphi}$$

POLI MULTIPLI COMPLESSI

$$F(s) = \frac{r_k^m}{(s-p)^m} + \frac{r_k^{m-1}}{(s-p)^{m-1}} + \dots + \frac{r_k}{(s-p)}$$

$$f(t) = \frac{t^{(k-1)}}{(k-1)!} (R e^{p t} + R^* e^{p^* t})$$

MODI PROPRI DEL SISTEMA

$$h, h e^{pt}, h e^{pt} \sin(\omega t + \varphi)$$

1 PAIR DI P.S.

$$Y(s) = \frac{h_1}{(s-p_1)} + \frac{h_2}{(s-p_1)} + \dots$$

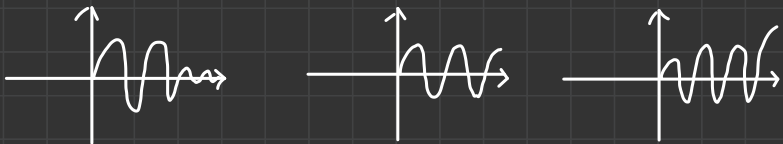
$$Y(s) = \frac{1}{s-p} \quad L^{-1} \left\{ \frac{1}{s-p} \right\} = h e^{pt} \quad y(0) = h_p$$

MODO OSCILLANTE

$$Y(s) = \frac{h}{(s-p)} + \frac{h^*}{(s-p^*)} \quad h = |h| e^{j\varphi}$$

$$y(t) = |h| e^{j\varphi} e^{(\sigma + j\omega)t} + |h| e^{-j\varphi} e^{(\sigma - j\omega)t}$$

$$y(t) = 2|h| e^{\sigma t} \cos(\omega t + \varphi)$$



RISPOSTA LIBERA E RISPOSTA FORZATA

$$y^{(m)} + a_{m-1} y^{(m-1)} + \dots = b_0 u(t) + \dots + b_m u^{(m)}(t)$$

$$\mathcal{L} \left\{ \frac{d^n}{dt^n} y(t) \right\} = s^n Y(s) + \mathcal{L} \{ y^{(m-1)}(s) + a_{m-1} s^{m-1} y(s) \dots$$

$$Y(s) = \frac{b_m s^m + \dots + b_0}{a_m s^m + \dots + a_0} U(s) - \frac{\mathcal{L} \{ y(s) \}}{a_m s^m + \dots + a_0}$$

$$Y(s) = Y_f(s) + Y_L(s) = F(s)U(s) + Y_L(s) = F(s)U(s)$$

RISPOSTA TRANSITORIA E RISPOSTA A REGIME PERMANENTE

$$F(s) = \frac{N(s)}{D(s)} \quad u(t) = \frac{t^k}{k!}$$

$$Y(s) = \frac{N(s)}{D(s)} U(s)$$

$$Y(s) = \sum_{k=0}^{m_{0,s}} \frac{p_k}{(s-p_k)} + \sum_{i=0}^{m_{s,s}} \frac{Q_i}{(s-p_i)} + \sum_{r=0}^{m_{\text{NSTR}}} \frac{R_i}{(s-p_i)}$$

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} s Y(s)$$

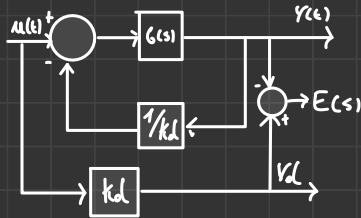
$$a_m s^m + \dots + a_0 = (s-p)(s-p)^* = (s-\sigma-j\omega)(s-\sigma+j\omega) = s^2 + \omega^2 - 2\sigma s + \sigma^2 = s^2 + 2\zeta\omega_n s + \omega_n^2$$

$$s^2 = \omega^2 \quad s = \pm j\omega$$

LUOGO DELLE RADICI

$$F(s) = \frac{k_p P(s)}{1 + k_p P(s)} = \frac{k_p \frac{N(s)}{D(s)}}{1 + k_p \frac{N(s)}{D(s)}} = \frac{k_p N(s)}{D(s) + k_p N(s)}$$

COMPORTAMENTO A REGIME



$$e = y_d - y \quad e(\infty) = 0$$

$$W(s) = \frac{G(s)}{1 + \frac{G(s)}{k_d}}$$

$$E(s) = k_d U(s) - W(s) U(s) = k_d U(s) - \frac{G(s)}{1 + \frac{G(s)}{k_d}} U(s) = \left(k_d - \frac{k_d G(s)}{k_d + G(s)} \right) U(s) = \frac{k_d^2}{k_d + G(s)} U(s)$$

$$u(t) = \frac{t^k}{k!} \quad U(s) = \frac{1}{s^{k+1}}$$

$$\lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s \cdot \frac{k_d^2}{k_d + G(s)} \cdot \frac{1}{s^{k+1}} = \lim_{s \rightarrow 0} \frac{k_d^2}{k_d + G'(s)} \cdot \frac{1}{s^h} \quad G(s) = C(s)P(s) \quad G'(s) = \frac{G'(s)}{s^h}$$

$$= \lim_{s \rightarrow 0} \frac{s^h k_d^2}{s^h k_d + G'(s)} \cdot \frac{1}{s^k} = \lim_{s \rightarrow 0} \frac{k_d^2}{s^h k_d + G'(s)} \cdot \frac{1}{s^{h-k}} \quad G'(0) = k_0$$

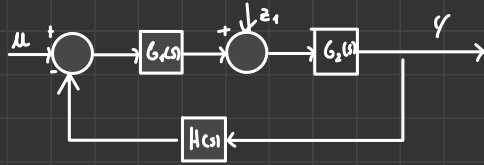
$$h > k \quad e(\infty) = 0$$

$$h < k \quad e(\infty) = \infty$$

$$h = k \quad \begin{matrix} h=1 & \frac{k_d^2}{k_g} \\ h=0 & \frac{k_d^2}{k_d + k_g} \end{matrix}$$

$h \setminus k$	$\sigma_1(t)$ $\frac{k_d^2}{k_d + k_g}$	$\sigma_2(t)$ $\frac{k_d^2}{k_g}$	$\sigma_3(t)$ $\frac{k_d^2}{k_d + k_g}$
0	$\frac{k_d^2}{k_d + k_g}$	∞	∞
1	0	$\frac{k_d^2}{k_g}$	∞
2	0	0	$\frac{k_d^2}{k_d + k_g}$

REIEZIONE DEI DISTURBI



$$W_2(s) = \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)} \quad Z(s) = \frac{1}{s^{k+1}}$$

$$\lim_{s \rightarrow 0} s W_2(s) Z(s) = \lim_{s \rightarrow 0} s \cdot \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)} \cdot \frac{1}{s^{k+1}} = \lim_{s \rightarrow 0} \frac{\frac{G_2(s)}{1 + \frac{G_1(s)}{s^h} G_2(s)H(s)}}{s^k} = \lim_{s \rightarrow 0} \frac{\frac{G_2(s)}{s^h + G_1(s)G_2(s)H(s)}}{s^{h-k}} \cdot \frac{1}{s^{h-k}}$$

$$G_1(s) = \frac{G_1(s)}{s^h}$$

$$G_1(0) = k_{G_1}$$

$$G_2(0) = k_{G_2}$$

$$h > k \quad e_z = 0$$

$$h < k \quad e_z = \infty$$

$$h = k \quad e_z = \frac{k_{G_2}}{k_{G_1} k_{G_2}} \frac{1}{k_d}$$

RISPOSTA ARMONICA

$$u(t) \rightarrow \boxed{g(t)} \rightarrow v(t) \quad u(t) = \sin(\omega t) \quad U(s) = \frac{\omega}{s^2 + \omega^2}$$

$$Y(s) = G(s) \frac{\omega}{s^2 + \omega^2}$$

$$Y(s) = \frac{r_1}{s - p_1} + \frac{r_2}{s - p_2} + \dots + \frac{r}{s - j\omega} + \frac{r^*}{s + j\omega}$$

$$r = \lim_{s \rightarrow j\omega} (s - j\omega) G(s) \frac{\omega}{(s - j\omega)(s + j\omega)} = \frac{G(j\omega)}{2j} \quad r^* = \frac{G^*(j\omega)}{-2j}$$

$$\left[-1 \sum \frac{r}{s - j\omega} \right] = r e^{j\omega t} \quad G(j\omega) = |G(j\omega)| e^{j\varphi}$$

$$y_p(t) = \frac{|G(j\omega)| e^{j\varphi}}{2j} e^{j\omega t} + \frac{|G(j\omega)| e^{-j\varphi}}{-2j} e^{-j\omega t} = |G(j\omega)| \sin(\omega t + \varphi)$$

DIAGRAMMI DI BODE

$$G(s) = K_G s^h \frac{\left(\frac{s^2}{\omega_n^2} + \frac{2\zeta}{\omega_n} s + 1 \right) (s\tau + 1)}{\left(\frac{s^2}{\omega_n^2} + \frac{2\zeta}{\omega_n} s + 1 \right) (s\tau + 1)}$$

GUADAGNO COSTANTE

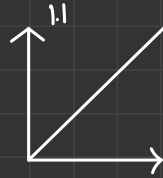
$$|K_G|_{dB} = 20 \log_{10} |K_G|$$

$$\angle K_G \quad \begin{array}{ll} K_G > 0 & 0^\circ \\ K_G < 0 & -180^\circ \end{array}$$

ZERI E POLI: NEW ORIGINI

$$S^h \xrightarrow{S=j\omega} (j\omega)^h$$

$$\lambda = \log_{10} \omega \quad \omega = 10\lambda$$



$$|(j\omega)^h|_{dB} = 20h \log_{10} \omega = 20h\lambda$$

$$\angle (j\omega)^h = 90h^\circ$$



ZERI E POLI: REALI

$$(s\tau + 1) \xrightarrow{S=j\omega} (j\omega\tau + 1)$$

$$|(j\omega\tau + 1)|_{dB} = \sqrt{(j\omega\tau)^2 + 1} \quad dB = 10 \log_{10} ((\omega\tau)^2 + 1)$$

$$\omega\tau \gg 1 \quad 20 \log_{10} (\omega\tau) = 20 \log_{10} \omega + 20 \log_{10} \tau \sim 20 \log_{10} \omega = 20\lambda$$

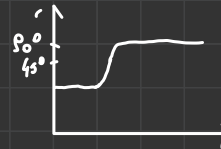
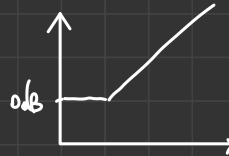
$$\omega\tau \ll 1 \quad |1| = 0 \text{ dB}$$

$$\omega\tau \approx 1$$

$$\omega\tau \gg 1 \quad \angle(j\omega\tau) = 90^\circ$$

$$\omega\tau \ll 1 \quad \angle(1) = 0^\circ$$

$$\omega\tau \approx 1 \quad \angle(j\tau) = 45^\circ$$



ZERI E POLI CONIUGATI:

$$\left(\frac{z^2}{\omega_n^2} + \frac{2\zeta}{\omega_n} s + 1 \right) \xrightarrow{s=j\omega} \left(\frac{2\zeta\omega}{\omega_n} + 1 - \frac{\omega^2}{\omega_n^2} \right)$$

$$\left| \frac{2\zeta\omega}{\omega_n} + 1 - \frac{\omega^2}{\omega_n^2} \right|_{dB} = \sqrt{\left(\frac{2\zeta\omega}{\omega_n} \right)^2 + \left(1 - \frac{\omega^2}{\omega_n^2} \right)^2} \quad |_{dB} = 10 \log_{10} \left(\left(\frac{2\zeta\omega}{\omega_n} \right)^2 + \left(1 - \frac{\omega^2}{\omega_n^2} \right)^2 \right)$$

$$\omega \gg \omega_n$$

$$40 \log_{10} \left(\frac{\omega}{\omega_n} \right) = 40 \log_{10} \omega - 40 \log_{10} \omega_n \sim 40 \log_{10} \omega = 40\lambda$$

$$\omega \ll \omega_n$$

$$|1|_{dB} = 0$$

$$\omega \approx \omega_n$$

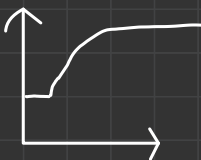
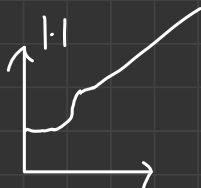
$$(2\zeta)_{dB} = 20 \log_{10} (2\zeta)$$

$$\angle \left(\frac{2\zeta\omega}{\omega_n} + 1 - \frac{\omega^2}{\omega_n^2} \right)$$

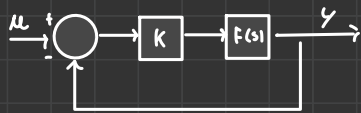
$$\omega \gg \omega_n \quad \angle \left(-\frac{\omega^2}{\omega_n^2} \right) = 180^\circ$$

$$\omega \ll \omega_n \quad \angle(1) = 0^\circ$$

$$\omega \approx \omega_n \quad \angle(j+1) = 45^\circ$$



SISTEMI A FASE NON MINIMA



$$F(s) = \frac{N(s)}{D(s)}$$

$$W(s) = \frac{k \cdot F(s)}{1 + k \cdot F(s)} = \frac{k \cdot \frac{N(s)}{D(s)}}{1 + k \cdot \frac{N(s)}{D(s)}} = \frac{k \cdot N(s)}{D(s) + k \cdot N(s)}$$

$$k \rightarrow 0$$

$$k \rightarrow \infty$$

SISTEMI CON RITARDO FINITO

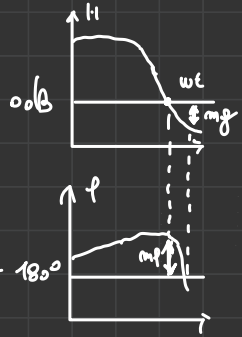
$$\mathcal{L}\{f(t-T)\} = F(s)e^{-sT}$$

$$s = j\omega$$

$$e^{-j\omega T}$$

MARGINI DI STABILITA'

$$Y(s) = F(s) U(s)$$

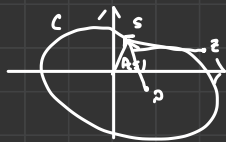


$$|F(j\omega)| = 1 \quad \angle F(j\omega) = -180$$

$$u(t) = \sin \omega t$$

C RITERIO DI NYQUIST

$$P(s) = k \frac{(s-z)}{(s-p)}$$



$$m^{\circ} \text{ZER}_{\text{loc}}(P(s)) - m^{\circ} \text{POL}_{\text{loc}}(P(s))$$



$$F(s) = \frac{N(s)}{D(s)}$$

$$W(s) = \frac{F(s)}{1+F(s)} = \frac{\frac{N(s)}{D(s)}}{1 + \frac{N(s)}{D(s)}} = \frac{N(s)}{D(s) + N(s)}$$

$$1+F(s) = 1 + \frac{N(s)}{D(s)} = \frac{D(s) + N(s)}{D(s)}$$

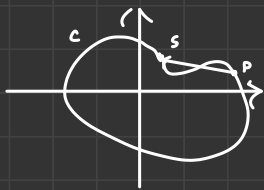


$$R_{(1+F(s)),0} = m^{\circ} \text{ZER}_{\text{PAP}}(1+F(s)) - m^{\circ} \text{POL}_{\text{PAP}}(1+F(s))$$

$$= m^{\circ} \text{POL}_{\text{PAP}}(W(s)) - m^{\circ} \text{POL}_{\text{PAP}}(F(s))$$

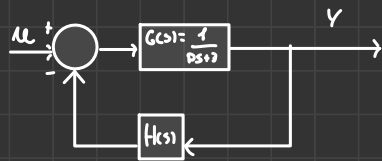
$$R_{(1+F(s)),0} = m^{\circ} \text{POL}_{\text{PAP}}(F(s))$$

$$R_{(F(s)),\infty} = -m^{\circ} \text{POL}_{\text{PAP}}(F(s))$$



FUNZIONE DI SENSIBILITA'

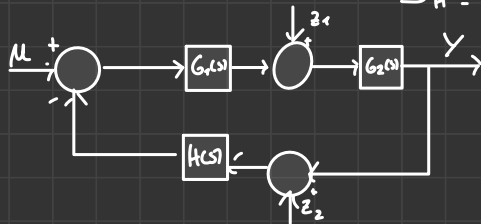
$$S_{\alpha}^Q = \frac{\frac{dQ}{d\alpha}}{\frac{dQ}{d\alpha}} = \frac{\alpha}{Q} \frac{dQ}{d\alpha}$$



$$S_p^W = \frac{P}{W} \cdot \frac{dW}{dP} = \left(\frac{G}{G} \frac{dG}{dG} \right) \left(\frac{P}{W} \frac{dW}{dP} \right) = \left(\frac{P}{G} \frac{dG}{dP} \right) \left(\frac{G}{W} \frac{dW}{dG} \right) = S_P^G \cdot S_G^W$$

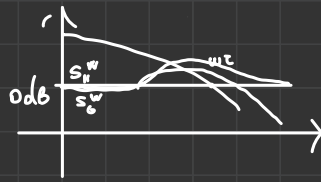
$$S_G^W = \frac{G(s)}{W(s)} \frac{dW(s)}{dG(s)} = \frac{1}{1 + G(s)H(s)}$$

$$S_H^W = \frac{H(s)}{W(s)} \frac{dW(s)}{dH(s)} = \frac{G(s)H(s)}{1 + G(s)H(s)}$$



$$Y(s) = \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)} Z_1(s) = S_G^W \cdot G_2(s) \cdot Z_1(s)$$

$$Y(s) = \frac{H(s)G_1(s)G_2(s)}{1 + G_1(s)G_2(s)H(s)} Z_2(s) = S_H^W \cdot Z_2(s)$$

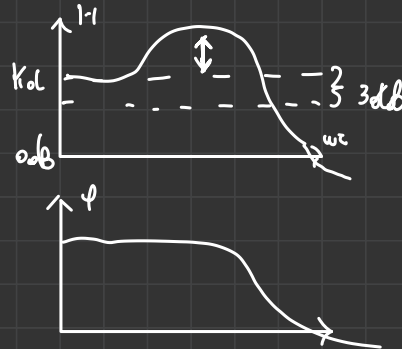
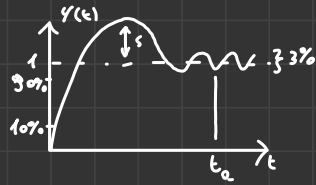


SPECIFICHE AD ANELLO CHIUSO E AD ANELLO APERTO PER UN SISTEMA DI CONTROLLO

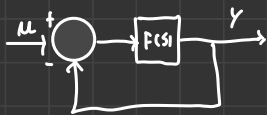


$$W(s) = \frac{F(s)}{1 + F(s)}$$

$$F(s) = \frac{W(s)}{1 - W(s)}$$

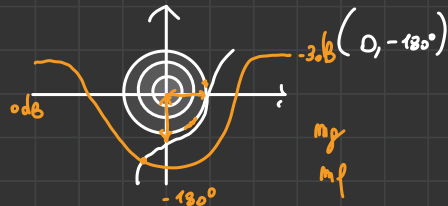


CARTE DI NICHOLS

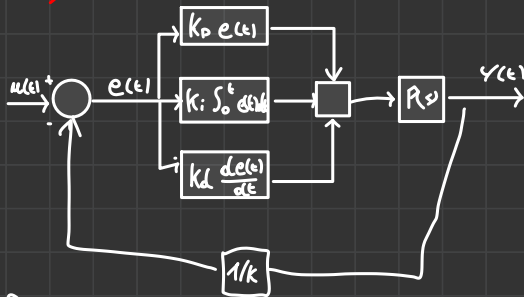


$$W(s) = \frac{F(s)}{1 + F(s)}$$

$$F(s) = \frac{W(s)}{1 - W(s)}$$



PID



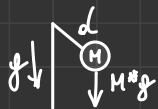
$$PID(s) = k_d s + \frac{k_i}{s} + k_p = \frac{k_d s^2 + k_p s + k_i}{s}$$

$$W(s) = \frac{k_p P(s)}{1 + k_p P(s)}$$

$$W(s) = \frac{k_i P(s)}{s + k_i P(s)}$$

LINEARIZZAZIONE INTORNO A UN PUNTO D'EQUILIBRIO

$$y = \sin(x) = \sin(x_0) + \frac{d(\sin(x_0))}{dx_0} (x - x_0) + \dots$$



$$M l^2 \ddot{\theta} = -D \dot{\theta} - M g l \sin(\theta) + \tau$$

$$M l^2 \ddot{\theta} + D \dot{\theta} + M g l \sin(\theta) = \tau$$

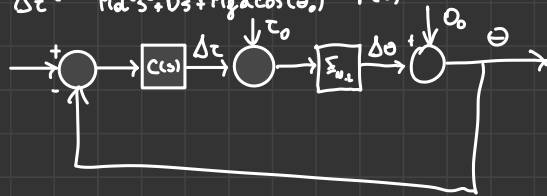
$$\theta = \theta_0 + \Delta\theta$$

$$\ddot{\theta} = \Delta\ddot{\theta} \quad \dot{\theta} = \Delta\dot{\theta} \quad \sin(\theta) = \sin(\theta_0) + \cos(\theta_0)\Delta\theta$$

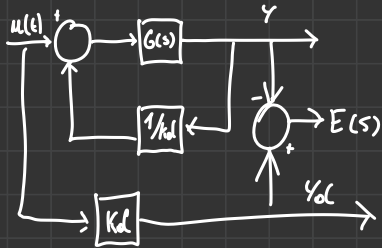
$$M l^2 \Delta\ddot{\theta} + D \Delta\dot{\theta} + M g l (\sin(\theta_0) + \cos(\theta_0)\Delta\theta) = \tau_0 + \Delta\tau \quad (\theta_0, \tau_0)$$

$$M g l \sin(\theta_0) = \tau_0 \quad \tau = \tau_0 + \Delta\tau$$

$$\frac{\Delta\theta}{\Delta\tau} = \frac{1}{M l^2 s^2 + D s + M g l \cos(\theta_0)} = P(s)$$



ERRORE DI RIPRODUZIONE DI UNA SINUSOIDE E RELEZIONE DEI DISTURBI SINUSOIDALI



$$E(s) = y_d(s) - y(s) = k_d U(s) - W(s) U(s) = \left(k_d - \frac{k_d G(s)}{k_d + G(s)} \right) U(s) = \frac{k_d^2}{k_d + G(s)} U(s)$$

$$W(s) = \frac{G(s)}{1 + \frac{G(s)}{k_d}} = \frac{k_d G(s)}{k_d + G(s)}$$

$$\frac{E(s)}{U(s)} = \frac{k_d^2}{k_d + G(s)} = \frac{k_d}{1 + \frac{G(s)}{k_d}} = \frac{k_d}{1 + F(s)} \sim \frac{k_d}{|F(s)|}$$

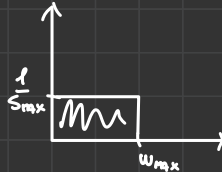
$$F(s) = \frac{G(s)}{k_d}$$

$$S_G^W = \frac{1}{1 + G(s)H(s)} = \frac{1}{1 + F(s)} \leq S_{max}$$

$$F(s) = G(s)H(s)$$

$$\frac{1}{|F(s)|} \leq S_{max}$$

$$|F(s\omega)| \geq \frac{1}{S_{max}}$$

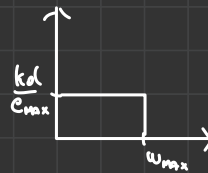


$$e \leq e_{max}$$

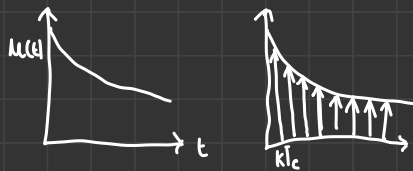
$$\omega \leq \omega_{max}$$

$$\frac{k_d}{|F(s\omega)|} \leq e_{max}$$

$$|F(s\omega)| \geq \frac{k_d}{e_{max}}$$



TEOREMA DEL CAMPIONAMENTO



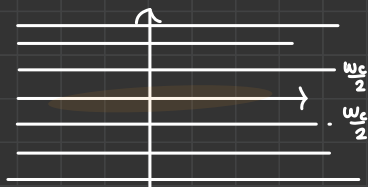
$$x_c(t) = \sum_{k=0}^{\infty} x(kT_c) \delta(t - kT_c)$$

$$\omega_c = 2\pi f_c \quad f_c = \frac{1}{T_c}$$

$$X_c(s) = \sum_{k=0}^{\infty} x(kT_c) e^{-skT_c}$$

$$T_c = 2\pi / \omega_c \quad s = \sigma + j\omega$$

$$= \sum_{k=0}^{\infty} x(kT_c) e^{-\sigma k \frac{2\pi}{\omega_c}} e^{-j\omega k \frac{2\pi}{\omega_c}}$$



$$f(t) = \frac{a_0}{2} \sum_{k=-\infty}^{\infty} [a_k \cos(\omega_c k t) + b_k \sin(\omega_c k t)]$$

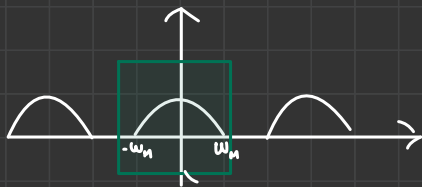
$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{-j\omega_c k t}$$

$$c_k = \frac{1}{T_c} \int_{-\frac{T_c}{2}}^{\frac{T_c}{2}} f(t) e^{-j\omega_c k t} dt = \frac{1}{T_c}$$

$$x_{c, \text{periodica}}(t) = \frac{1}{T_c} \sum_{k=-\infty}^{\infty} e^{-j\omega_c k t}$$

$$x_c(t) = x(t) \cdot \frac{1}{T_c} \sum_{k=-\infty}^{\infty} e^{-j\omega_c k t} = \frac{1}{T_c} \sum_{k=-\infty}^{\infty} x(t) e^{-j\omega_c k t}$$

$$X_c(s) = \frac{1}{T_c} \sum_{k=-\infty}^{\infty} X(s - j\omega_c k)$$



$k\omega_c$

ORGANO DI TESSUTO



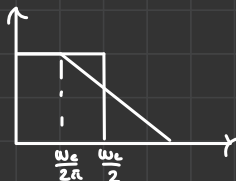
$$x_{\text{GABINO}}(t) = \sum_{k=0}^{\infty} x_k \delta(t - kT_c) - x_k \delta(t - (k+1)T_c)$$

$$X_{\text{GABINO}}(s) = \sum_{k=0}^{\infty} x_k e^{-skT_c} - x_k e^{-s(k+1)T_c} = \sum_{k=0}^{\infty} x_k \frac{1}{s} e^{-skT_c} - x_k \frac{1}{s} e^{-s(k+1)T_c} = \left(\sum_{k=0}^{\infty} x_k e^{-skT_c} \right) \left(\frac{1 - e^{-sT_c}}{s} \right)$$

$$T(s) = \frac{1 - e^{-sT_c}}{s} = \frac{1}{s} \cdot 1 - \frac{1}{s} e^{-sT_c} = \frac{1}{s} \cdot \frac{1 + sT_c - 1}{1 + sT_c} = \frac{T_c}{1 + sT_c}$$

$$e^x = 1 + x$$

$$e^{sT_c} = 1 + sT_c$$



SCELTA DEL TEMPO DI CAMPIONAMENTO

$$\omega_c > 2\omega_{max} \quad 2\pi \frac{1}{T_c} > 2\omega_{max} \rightarrow \frac{1}{T_c} > \frac{\omega_{max}}{\pi} \quad T_c > \frac{\pi}{\omega_{max}}$$

$$\omega_c = 2\pi \frac{1}{T_c}$$

TRASFORMATA Z

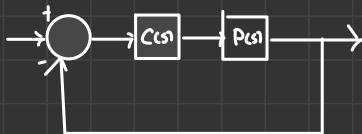
$$y_{k+1} = ay_k + u_k$$

$$X(z) = \sum_{k=0}^{\infty} x_k z^{-k} = \frac{1}{1 - x_k z^{-1}} \quad |x_k z^{-1}| < 1$$

$$X_c(s) = \sum_{k=0}^{\infty} x_k e^{-skT_c} \quad z = e^{sT_c}$$

$$X_c(z) = \sum_{k=0}^{\infty} x_k z^{-k}$$

TRASFORMATA ESATA



$$C(s) = \frac{1}{s-p}$$

$$c(t) = e^{pt}$$

$$c_e(k) = \sum_{k=0}^{\infty} e^{pkT_c} \delta(t - kT_c)$$

$$C(z) = \sum_{k=0}^{\infty} e^{pkT_c} z^{-k} = \sum_{k=0}^{\infty} (e^{pT_c} z^{-1})^k = \frac{z}{1 - e^{pT_c} z^{-1}} \quad |e^{pT_c} z^{-1}| < 1 \quad |z| > e^{pT_c}$$

$$\lim_{s \rightarrow 0} s \cdot \frac{1}{s-p} \cdot \frac{1}{s} = -\frac{1}{p}$$

$$\lim_{z \rightarrow 1} (1/z^c) \cdot \frac{z}{1-e^{0ic}} \cdot \frac{1}{(1-z)^c} = \frac{1}{1-e^{0ic}}$$

MAPPING DAL PIANO S AL PIANO Z

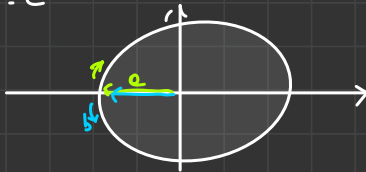
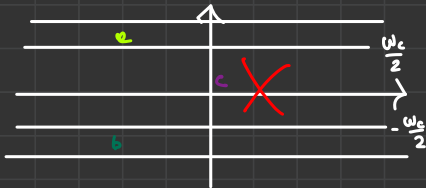
$$X_c(s) = \sum_{k=0}^{\infty} x_k e^{-skT_c}$$

$$X_c(z) = \sum_{k=0}^{\infty} x_k z^{-k}$$

$$z = e^{sT_c} = e^{\sigma \frac{2\pi}{\omega_c}} \cdot e^{j\omega \frac{2\pi}{\omega_c}} = e^{s\omega \frac{2\pi}{\omega_c}}$$

$$e^{s \frac{\omega_c}{2} \frac{2\pi}{\omega_c}} = e^{s\pi}$$

$$= e^{-s\pi}$$



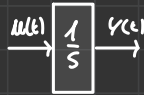
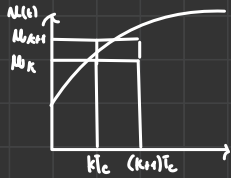
$$z = e^{s\omega \frac{2\pi}{\omega_c}} = 1$$

$$|z| = \sqrt{\sigma^2 + \omega^2}$$

$$|z| \geq 1$$

$$|z| < 1$$

T TRASFORMAZIONE APPROSSIMATA DAL PIANO S AL PIANO Z



1. Difetto

$$y_{k+1} = y_k + M_K T_c$$

$$Y(z)(1 - z^{-1}) = T_c U(z) z^{-1}$$

$$\frac{Y(z)}{U(z)} = \frac{T_c z^{-1}}{1 - z^{-1}} \approx \frac{1}{s} \rightarrow s = \frac{1 - z^{-1}}{T_c}$$

2. Eccesso

$$y_{k+1} = y_k + M_{K+1} T_c$$

$$Y(z)(1 - z^{-1}) = T_c U(z)$$

$$\frac{Y(z)}{U(z)} = \frac{T_c}{1 - z^{-1}} \approx \frac{1}{s} \rightarrow s = \frac{1 - z^{-1}}{T_c}$$

3. Forma Trappazoidale

$$y_{k+1} = y_k + (M_K + M_{K+1}) \frac{T_c}{2}$$

$$Y(z)(1 - z^{-1}) = \frac{T_c}{2} U(z) (1 + z^{-1})$$

$$\frac{Y(z)}{U(z)} = \frac{T_c}{2} \frac{1 + z^{-1}}{1 - z^{-1}} \approx \frac{1}{s} \rightarrow s = \frac{2}{T_c} \frac{1 - z^{-1}}{1 + z^{-1}}$$

